

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-DC-3401A-01 Sheet 1 of 1
OPL Title	Reduction DC-3401A - 4-Step Overview	Discipline	Process / Operations	
Equipment	DC-3401A - CATALYST REDUCTION REACTOR	Area / Unit	3400 - CATALYST REDUCTION SYSTEM	
Related Interlock	SEQ-3401 (SIL 2)	P&ID Ref	TJC-LLD-PID-3401	
Classification:	☒ Basic Knowledge	☐ Improvement	☐ Trouble Case	Aspect: P-Q-C-D-S-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **reduction dc-3401a - 4-step overview** for DC-3401A (CATALYST REDUCTION REACTOR) in the CATALYST REDUCTION SYSTEM. The catalyst reduction is done in four controlled steps: Inerting, Heating-Up, Reduction, then Cooling & Preservation. Skipping or rushing a step risks an H2/O2 explosion or ruined catalyst. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **N2 / H2 / Pd catalyst** at up to 6 barg / FV / 250 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Inerting: purge with N2 via FV-17343 until O2 below 100 ppm	Permit valid & briefed
2	Heating-Up: heater EA-3401 on, N2 at 275 kg/h, ramp TIC-17343 to 180-210 degC	Parameter within limit
3	Reduction: introduce H2, hold about 12 hours (PdO + H2 -> Pd)	Interlock status confirmed
4	Cooling: cool all bed below 29 degC, then hold under N2 at 2 kg/cm2	Change logged on DCS
5	Log every step transition and confirm permissives before the next step	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Bed multipoint TC point 3 reading open circuit	Thermocouple element failed at the sheath junction	Replaced TC point 3 in the multipoint assembly, verified reading.
One heater bank drawing no current, slow heat-up	Electric heater element bank open circuit / element burnt	Replaced heater element bank, meggered, verified TIC-17343 response
Inerting O2 analyzer drift vs portable reference	Analyzer zero drift and sample dryer saturated	Recalibrated AI-3401 on span gas, replaced sample dryer

6. KEY LEARNING POINTS

- The catalyst reduction is done in four controlled steps: Inerting, Heating-Up, Reduction, then Cooling & Preservation. Skipping or rushing a step risks an H2/O2 explosion or ruined catalyst.
- Respect the interlock SEQ-3401: e.g. TSHH-3401 trips at > 230 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-3401 before working on DC-3401A.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
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